

ArthRakshak

A Financial Digital Twin for Every Indian Household

Not just advice. A guardian that understands your life,
predicts your next financial stress, and guides you
before you make a costly mistake.

Financial
Digital Twin

Life-Event
Simulator

Scam
Shield

Behavioral
Engine

Team FinTwin Labs

Abhisheksingh Dhangar | **Mohd Bilal Khan**

Full-Stack & AI Engineer | AI Engineer & Backend Developer
IIIT Pune

React · Python · PyTorch · Deep Learning · FastAPI · LangGraph

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1 Executive Summary

💡 The Core Insight

India’s financial inclusion gap is no longer about **access** — UPI processes 200+ billion transactions annually with 504 million unique users. The gap is about **understanding**. No existing solution is built for how India’s most vulnerable populations *actually live*: irregular incomes, local languages, low literacy, disability, and pervasive scam exposure.

ArthRakshak (*Arth* = Wealth, *Rakshak* = Guardian) is an **agentic AI system** that acts as a personal financial guardian for every Indian household. Unlike conventional budgeting tools or chatbots, ArthRakshak:

- 1. **Builds a Financial Digital Twin** — a living, evolving virtual model of each user’s income variability, spending behavior, debt exposure, savings capacity, goal timelines, risk tolerance, and vulnerability profile.
- 2. **Simulates Life Events** — shows users the real consequences of “what-if” scenarios (income drops, medical emergencies, loan EMIs, crop failure) *before* they happen.
- 3. **Deploys a Scam Shield** — a central defense pillar that detects predatory loan apps, Ponzi schemes, phishing messages, and pressure-tactic manipulation in real time.
- 4. **Applies Behavioral Finance** — uses nudge theory, commitment devices, micro-goals, and contextual alerts to change behavior, not just dispense advice.
- 5. **Speaks Every Language, Serves Every Ability** — multilingual (Hindi, Marathi, Kannada, Tamil, and more), multimodal (voice, text, image/screenshot), and accessibility-first design for persons with disabilities.

12M+

Gig workers in India (FY2025)

40%

Earn < ₹15,000/month

80%

Rural women lack digital financial literacy

₹ 48K Cr

Financial fraud losses (FY2026)

2 Problem Statement: The Understanding Gap

2.1 Beyond Access — The Real Crisis

India has achieved remarkable success in financial access infrastructure. The Jan Dhan–Aadhaar–Mobile (JAM) trinity, combined with UPI, has created the world’s largest real-time payments ecosystem. The RBI’s Financial Inclusion Index reached **67.0 in March 2025**, up from 64.2 a year prior.

Yet access has not translated into understanding:

⚠ The Paradox of Digital Finance

While UPI accounts for **90% of retail payment volumes**, studies show only **27% average financial literacy** among adults. The gap between *transacting digitally* and *understanding financial decisions* has never been wider.

2.2 Why Current Solutions Fail

Table 1: Limitations of Existing Financial Tools

Existing Approach	Critical Limitation	Fails For
Budgeting Apps	Assume fixed monthly salary; no irregular income modeling	Rajesh
Bank Advisors	Push commission-driven products (ULIPs, endowment plans)	Priya
YouTube / Social Media	Unvetted content; financial influencer scams rampant	All
Govt. Scheme Portals	Complex forms; language barriers; no personalization	Kisan
Banking Apps	Not screen-reader friendly; visual-heavy interfaces	Divya
Generic AI Chatbots	Reactive; no context memory; no life-event modeling	All

2.3 The Four Faces of Financial Exclusion

India's financial literacy crisis affects distinct populations differently. ArthRakshak is designed around four representative personas derived from the hackathon brief:

👤 Priya — The Salaried Professional

Profile: 32-year-old school teacher in Thane, earning ₹35,000/month.

Pain Points:

- Mis-sold ULIPs and complex mutual funds by her bank every visit
- WhatsApp flooded with unverified “investment tips” she cannot evaluate
- Wants to save for daughter's education & retirement but lacks trusted guidance
- Financial articles are either too technical or irrelevant to her income level
- Has missed employer benefits and paid unnecessary charges unknowingly

Core Need: Clarity, trust, and protection from mis-sold products — in Marathi.

👤 Rajesh — The Gig Worker

Profile: 24-year-old food delivery & freelance worker in Mumbai, earning ₹15,000–40,000/month.

Pain Points:

- Income swings make traditional budgeting impossible
- Banks refuse services: “Come back when you have ₹5 lakhs”
- No emergency fund, no health insurance, frequent debt trap cycles
- Falls prey to instant loan apps with predatory interest rates
- YouTube finance content designed for salaried professionals is irrelevant

Core Need: Survival-level cashflow planning with irregular income support.

9 Kisan — The Agricultural Worker

Profile: 45-year-old farmer in rural Karnataka, cultivating 3 acres of land.

Pain Points:

- Seasonal, weather-dependent income creates massive cash flow gaps
- Takes high-interest informal loans during sowing season
- Knows about PM-KISAN, crop insurance — but application processes are confusing
- Loses money to middlemen; has no access to market price information
- Financial services not designed for agricultural income patterns in Kannada

Core Need: Seasonal planning, scheme navigation, and voice-first local language interaction.

9 Divya — The Visually Impaired Professional

Profile: 28-year-old content writer in Pune with visual impairment, earning ₹25,000/month.

Pain Points:

- Banking apps and financial platforms are not screen-reader friendly
- Charts, graphs, and visual financial content are inaccessible
- Relies on family members for complex financial tasks — compromising privacy
- Wants to learn about EPF, tax-saving instruments, and investing independently
- Government disability schemes exist but are hard to discover and navigate

Core Need: Accessibility-first interaction — voice, audio, and non-visual guidance.

3 Solution: ArthRakshak — The Financial Guardian System

3.1 Design Philosophy

ArthRakshak is built on three foundational principles that differentiate it from every existing solution:

1. **Proactive, Not Reactive:** Most tools wait for questions. ArthRakshak's Financial Digital Twin continuously monitors and *initiates* guidance: “*Rajesh, this week you are safe to spend only ₹1,800*” or “*Priya, your recurring monthly leak suggests a SIP of ₹2,000 is safe.*”
2. **Experience-Based Learning:** People do not learn from definitions. The Life-Event Simulator transforms financial literacy into *lived experience*: “*What happens if your income drops 40% next month?*” — showing real consequences with the user's own data.
3. **Trust Through Community:** Financial trust is not built by AI alone. ArthRakshak integrates a **Human-AI-Community Loop** — NGO/educator verification, community ambassador onboarding, and human escalation for sensitive decisions.

3.2 Core Modules

Table 2: ArthRakshak Core Module Architecture

Module	Description
Financial Twin Engine	Builds and maintains a living virtual model of each user’s income, expenses, goals, risk tolerance, debt exposure, savings capacity, language preference, confidence level, and scam vulnerability. Continuously updated with real transaction data.
Life-Event Simulator	Interactive “what-if” engine: income drops, medical bills, loan EMIs, school fees, crop failure, better savings paths. Turns abstract financial concepts into tangible, personalized outcomes.
Cashflow Agent	Handles irregular income modeling for gig workers and farmers. Computes weekly safe-to-spend amounts, identifies savings windows, and predicts debt trap risk from spending patterns.
Goal Planner Agent	Creates personalized SIP, emergency fund, insurance, retirement, and debt repayment plans. Adapts goal timelines dynamically based on actual income data — not assumed fixed salaries.
Education Agent	Explains credit scores, mutual funds, insurance, ULIPs, tax-saving instruments in plain Hindi, Marathi, Kannada, Tamil, and other regional languages. Uses storytelling, analogies, and bite-sized lessons.
Scam Shield Agent	Central defense pillar: detects fake loan apps, Ponzi schemes, phishing SMS, WhatsApp investment forwards, predatory lending language, and pressure tactics. Assigns risk scores: Safe / Suspicious / High Risk / Needs Human Review.
Accessibility Agent	Adapts interaction mode: voice-first for Divya, simple text for Kisan, visual dashboards for Priya. Screen-reader compatible. Offline cache for low-connectivity areas.
Policy & Safety Agent	Validates all AI recommendations for accuracy, regulatory compliance, and user safety. Prevents harmful or misleading financial advice. Confidence scoring on every output.
Human Escalation Agent	Routes edge cases, high-risk situations, and sensitive decisions to verified human advisors — NGO counselors, certified financial planners, or community ambassadors.
Behavior Engine	Nudge system powered by behavioral finance: commitment devices, micro-goals, spending cooldown prompts, “safe-to-spend” alerts, savings streaks, milestone celebrations, and social reinforcement with privacy protection.

3.3 Financial Digital Twin — The Core Innovation

The Financial Digital Twin is ArthRakshak’s most distinctive technical contribution. For each user, the system builds and continuously updates a comprehensive virtual financial identity:

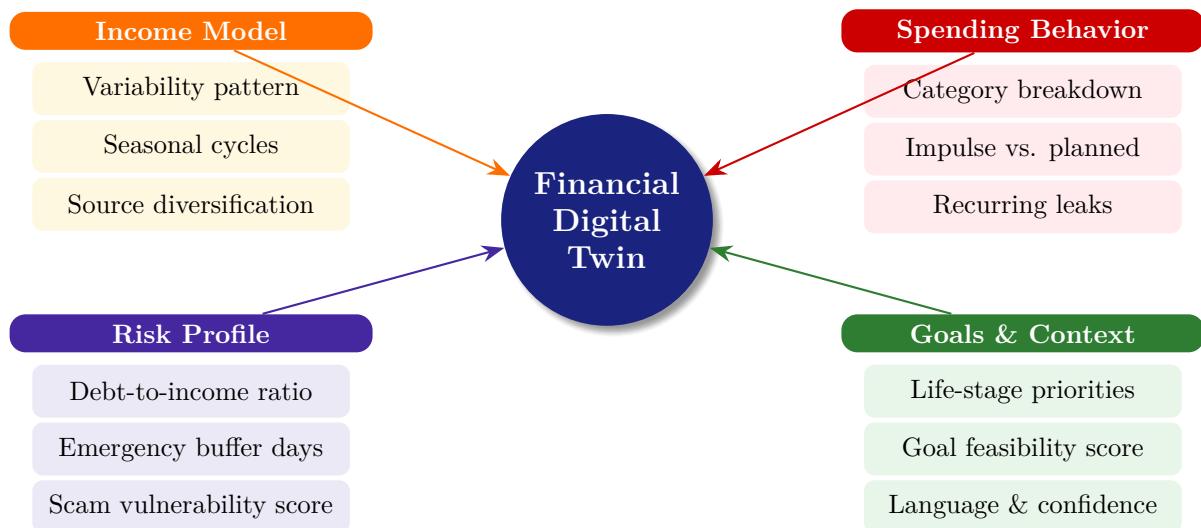


Figure 1: Financial Digital Twin — Multi-Dimensional User Model

✓ Why This Wins

Traditional finance apps are *reactive* — they answer when asked. The Financial Digital Twin makes ArthRakshak *anticipatory*:

- **Rajesh:** “This week you are safe to spend only ₹1,800. Save the rest for your EMI due on the 15th.”
- **Priya:** “Your recurring monthly leak of ₹3,200 on subscriptions suggests a SIP of ₹2,000 is safe — and you won’t feel it.”
- **Kisan:** “Your harvest income pattern shows a 3-month cash gap. Let me help you plan for it now.”
- **Divya:** “You’re eligible for a government disability scheme worth ₹6,000/year. I can guide you through the application via voice.”

3.4 Life-Event Simulator — Learning Through Experience

The Life-Event Simulator is ArthRakshak’s differentiator against every conventional finance app. It transforms financial literacy from *explanation* to *experience*.

Table 3: Life-Event Simulation Scenarios

Scenario	What the User Sees	Persona
Income Drop	“If your income drops 40% next month, you can survive 23 days before debt. Here’s how to extend that to 45 days.”	Rajesh
Medical Emergency	“A ₹50,000 medical bill today would wipe your emergency fund. With ₹200/month health insurance, you’d be covered.”	Priya
Loan EMI Start	“Adding a ₹5,000 EMI reduces your safe-to-spend by 35%. Consider this 6-month pre-savings plan first.”	All
Crop Failure	“If the monsoon delays 3 weeks, your cash reserve covers only sowing costs. Here’s a buffer plan.”	Kisan
Scam Encounter	“If you invest ₹10,000 in this ‘guaranteed 40% return’ scheme, here’s what historically happens to such schemes.”	All
Better Savings Path	“Switching from fixed deposit to a balanced mutual fund SIP could grow your ₹2,000/month by ₹1.8 lakh more over 10 years.”	Priya

💡 Pedagogical Foundation

The Life-Event Simulator is grounded in **experiential learning theory** (Kolb, 1984) and **prospect theory** (Kahneman & Tversky, 1979). Research shows people understand financial risk 4x better through simulated personal scenarios than through abstract explanations. By using the user’s *own* financial data, the simulator creates emotionally resonant learning moments that drive lasting behavioral change.

3.5 Scam Shield — A Central Defense Pillar

The Scam Shield is not a feature — it is a **core pillar** of ArthRakshak. With financial fraud losses reaching ₹48,021 crore in FY2026 and 87 unauthorized loan apps blocked by MeitY in late 2025, scam defense is mission-critical.

3.5.1 Detection Capabilities

Table 4: Scam Shield Detection Matrix

Threat Type	Detection Method	Risk Level
Fake Loan Apps	Cross-reference RBI DLA Directory; APK signature analysis	High Risk
Ponzi Schemes	“Guaranteed returns” language detection; historical pattern matching	High Risk
Phishing SMS/Links	URL reputation scoring; sender pattern analysis	Suspicious
WhatsApp Investment Tips	Forwarded message metadata; claim verification via RAG	Suspicious
Predatory Lending	APR extraction; hidden fee detection; comparison with RBI norms	High Risk
Pressure Tactics	Urgency language detection; artificial scarcity patterns	Suspicious
Mis-sold Products	Product suitability scoring against user’s Digital Twin profile	Caution

3.5.2 Risk Scoring System

Every financial input, message, or product encountered by the user receives a risk classification:



Human Escalation: High-risk and ambiguous cases are routed to verified financial counselors via the Human Escalation Agent, ensuring no critical decision relies solely on AI judgment.

4 Technical Architecture

4.1 Multi-Agent Orchestration with LangGraph

ArthRakshak employs a **hierarchical multi-agent architecture** built on LangGraph’s stateful graph-based orchestration. This is fundamentally different from linear chatbot pipelines — each agent is a specialized expert with persistent state, memory, and tool access.

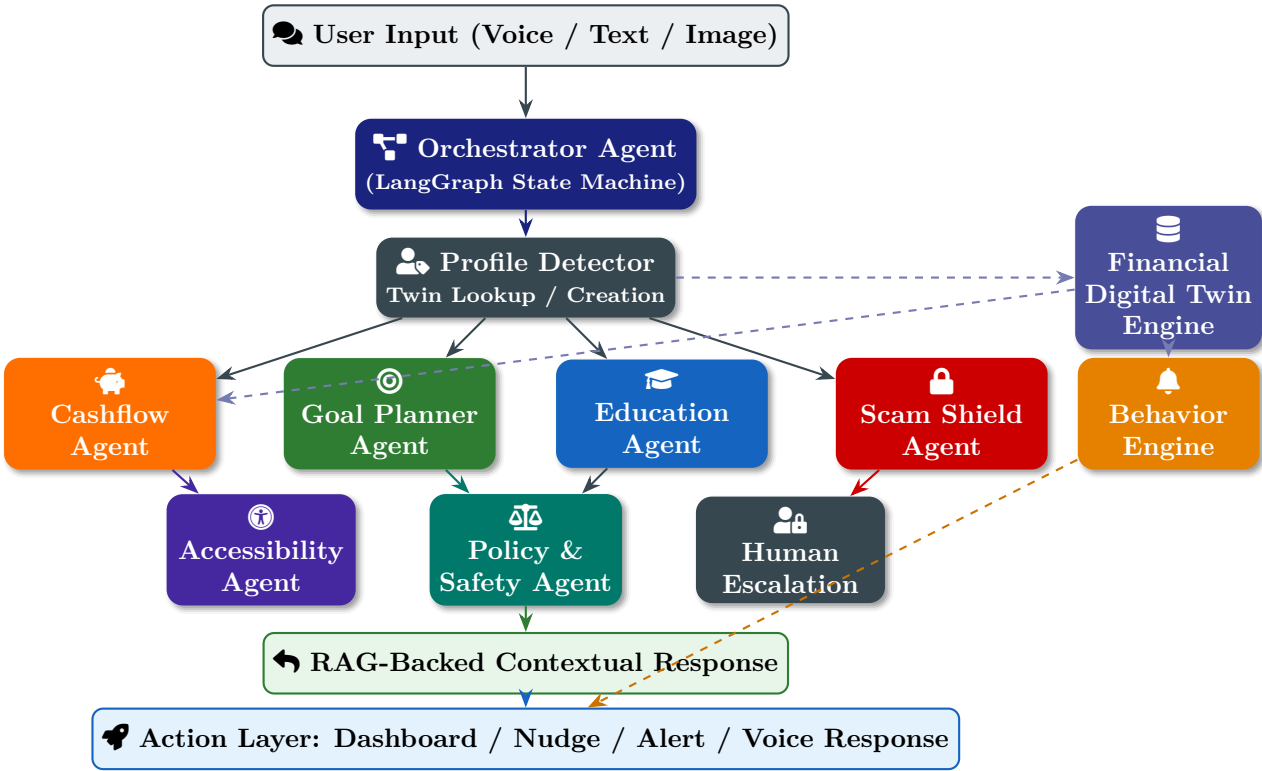


Figure 2: ArthRakshak Multi-Agent Architecture — Hierarchical Orchestration

4.2 Agent Communication Protocol

Table 5: Agent Routing Logic via LangGraph State Machine

User Intent	Routed Agent(s)	Example Query
Income tracking	Cashflow + Twin	“Yeh mahine kitna kharcha kar sakta hoon?”
Goal planning	Goal Planner + Twin	“Beti ki padhai ke liye kitna bachana chahiye?”
Concept explanation	Education + RAG	“Mutual fund kya hota hai? Simple mein batao”
Scam check	Scam Shield	“Yeh loan app safe hai kya?”
What-if simulation	Life-Event Sim + Twin	“Agar income 50% gir jaye toh kya hoga?”
Accessibility switch	Accessibility	“Voice mode mein baat karo”
High-risk decision	Human Escalation	“Should I take a ₹2 lakh personal loan?”

4.3 Technology Stack

Table 6: Production Technology Stack

Layer	Technology	Rationale
Frontend	React 18.2, TailwindCSS 3.4	Component-driven, responsive, accessible UI
Backend API	Python 3.11, FastAPI 0.110	Async-first, high-performance REST/WebSocket
Agent Framework	LangGraph 0.1.19, LangChain 0.1.20	Stateful multi-agent graphs with HITL support
LLM	Claude 3 Sonnet (Anthropic API)	Strong reasoning, safety alignment, multilingual
Database	PostgreSQL 16, SQLAlchemy 2.0	ACID-compliant, relational user/transaction data
Vector Store	ChromaDB 0.4.22 (local)	RAG for financial education content retrieval
Authentication	JWT via python-jose 3.3.0	Stateless, secure token-based auth
Speech / NLP	Whisper (voice input), gTTS	Multilingual voice-first interaction
Observability	LangSmith	Agent decision tracing and audit logging
Deployment	Docker, Docker Compose	Reproducible, containerized microservices

4.4 Data Flow Architecture

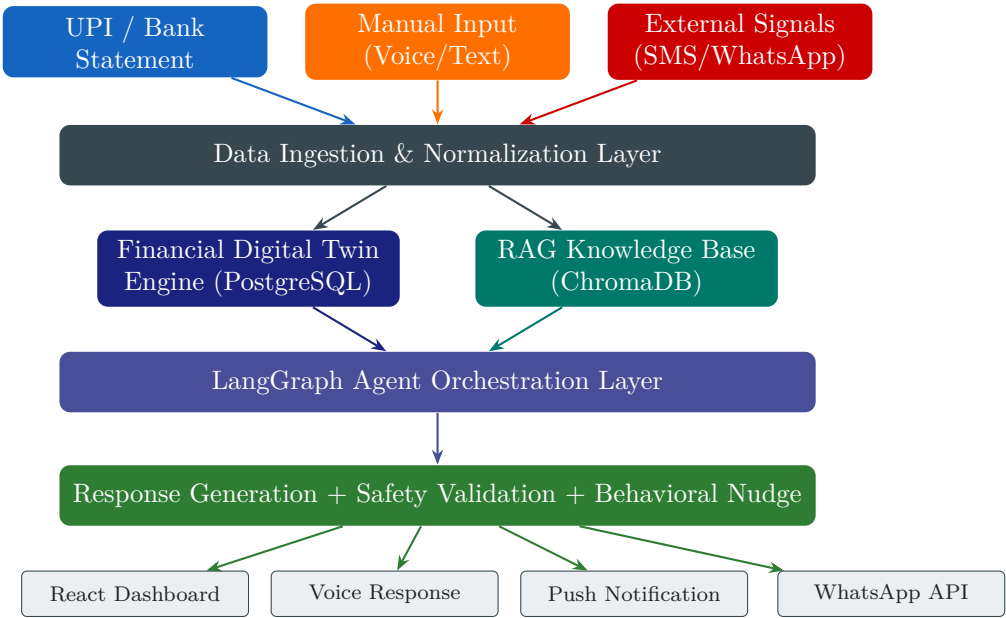


Figure 3: End-to-End Data Flow — From Input to Multimodal Delivery

5 Behavioral Finance Engine

ArthRakshak goes beyond information delivery. Grounded in **Nudge Theory** (Thaler & Sunstein, 2008) and **Behavioral Economics**, the Behavior Engine transforms the product from *advice-giving*

to *behavior-changing*.

5.1 Behavioral Mechanics

Table 7: Behavioral Finance Interventions

Mechanism	Implementation	Bias Addressed
Commitment Devices	Users set public savings goals; system enforces nudges at deviation	Present Bias
Micro-Goals	Break ₹ 50,000 emergency fund into “₹ 150/day for 11 months”	Goal Gradient
Spending Cooldown	24-hour delay prompt before large unplanned purchases	Impulse Bias
Safe-to-Spend Alerts	Daily proactive alert with remaining safe budget	Mental Accounting
Loss Framing	“Not saving ₹ 100 today costs you ₹ 18,000 over 10 years”	Loss Aversion
Savings Streaks	Visual streak counter; milestone celebrations at 7, 30, 90 days	Status Quo Bias
Social Proof	“83% of users like you started an SIP this month” (anonymized)	Herd Behavior
Default Enrollment	Auto-suggest optimal SIP amount based on Twin analysis	Default Effect
Confidence Building	Progressive complexity: simple concepts first, advanced later	Dunning-Kruger

💡 Ethical Nudging Commitment

ArthRakshak’s nudges are designed exclusively to serve the user’s long-term financial well-being. We explicitly commit to **never** using behavioral mechanics to:

- Push commission-driven financial products
- Create artificial urgency or fear to drive action
- Manipulate users into decisions that benefit the platform over the user
- Exploit cognitive biases for engagement metrics

All nudge algorithms are auditable and subject to the Policy & Safety Agent’s review.

6 Accessibility & Multimodal Interaction

6.1 Universal Design Principles

ArthRakshak is designed from the ground up to be **accessible to every Indian**, regardless of literacy, language, physical ability, or connectivity.

Table 8: Multimodal Interaction Capabilities

Modality	Capability	Primary Persona
Voice Input	Whisper-based speech recognition for Hindi, Marathi, Kannada, Tamil	Kisan, Divya
Voice Output	gTTS-based natural language response in user’s preferred language	All
Text Chat	Standard text interface with auto-language detection	Priya, Rajesh
Image/Screenshot	OCR + LLM analysis of forwarded messages, loan documents, bank statements	All
WhatsApp Integration	Business API for familiar, frictionless interaction	Kisan, Rajesh
Offline Cache	Critical financial rules and recent data cached locally	Rural users
Screen Reader	ARIA-compliant UI; all charts described in text/audio	Divya
Simple Language Mode	Flesch-Kincaid grade 5 equivalent; no jargon	Low literacy

6.2 Language Support Architecture



Language is auto-detected from user input and persisted in the Digital Twin profile. All education content, nudges, and alerts are delivered in the user’s preferred language with culturally appropriate analogies.

7 Responsible AI & Safety Framework

7.1 Safeguards for Vulnerable Users

⚠ Critical: Protecting the Most Vulnerable

ArthRakshak serves populations that are *most susceptible* to financial harm: low-income gig workers, elderly farmers, persons with disabilities, and individuals with low financial literacy. Our safety framework is not an afterthought — it is architecturally embedded.

- Policy Agent as Gatekeeper:** Every AI-generated recommendation passes through the Policy & Safety Agent before reaching the user. This agent:
 - Validates financial accuracy against SEBI/RBI/IRDAI regulations
 - Scores confidence level (Low / Medium / High) on every output
 - Flags recommendations that exceed the user’s risk profile
 - Blocks any advice that could be construed as “investment advice” without proper disclaimers
- Transparency in AI Limitations:**

- Every response includes a clear disclaimer: “I am an AI assistant. For important financial decisions, consult a certified financial planner.”
- Confidence scores are shown to the user when below threshold
- The system explicitly states when it “doesn’t know” rather than hallucinating

3. Human Escalation Protocol:

- Automatic escalation for: loans > ₹50,000, insurance decisions, tax filing, legal matters
- User can request human advisor at any time
- NGO/educator partnerships for ground-level trust building

4. Data Privacy & Security:

- All financial data encrypted at rest (AES-256) and in transit (TLS 1.3)
- No data shared with third parties without explicit consent
- User can delete all data at any time (right to be forgotten)
- Local-first processing where possible; cloud only for LLM inference

7.2 Audit Trail & Observability

Every agent interaction is logged via **LangSmith**, creating a complete decision audit trail:

- Which agent handled the query and why
- What tools were invoked and their outputs
- What reasoning chain led to the recommendation
- What safety checks were applied
- What confidence score was assigned

This is critical for regulatory compliance, user trust, and continuous system improvement.

8 Human-AI-Community Trust Loop

Financial trust is **not built by AI alone**. ArthRakshak integrates a three-layer trust architecture:

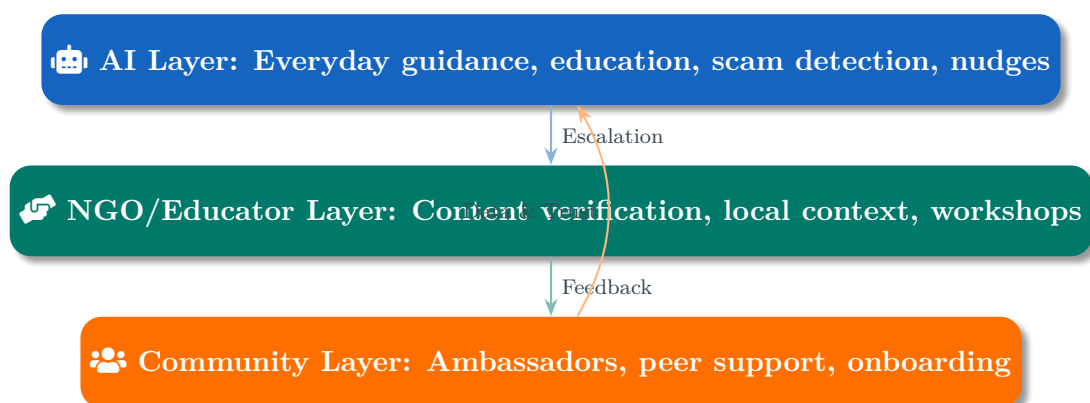


Figure 4: Human-AI-Community Trust Loop

- **AI Layer:** Handles 90% of everyday queries — budgeting, education, scam checks, nudges. Available 24/7 in local languages.
- **NGO/Educator Layer:** Trusted organizations (like SEWA, Pratham, local credit cooperatives) verify regional content, validate AI recommendations for cultural appropriateness, and provide ground-level training.
- **Community Ambassador Layer:** Trained local individuals who onboard new users, demonstrate the app, and provide peer-level trust that technology alone cannot create. Especially critical in rural areas.

9 Persona-Specific Journeys

9.1 Priya's Journey

Priya's ArthRakshak Experience

Day 1 — Onboarding: Priya opens ArthRakshak. Voice prompt in Marathi: “Namaste Priya! Mi tumcha arthik sahayak aahe. Tumhala kuthe madad havi?” She says she wants to save for her daughter's education. The system creates her Digital Twin.

Week 1 — Trust Building: The Education Agent explains what a ULIP actually is — in simple Marathi with a story analogy. Priya realizes her bank's product is wrong for her goals. She feels informed for the first time.

Month 1 — Scam Shield: Priya forwards a WhatsApp message promising “100% guaranteed returns.” The Scam Shield flags it as **High Risk** and explains why: no SEBI registration, impossible return claims, classic Ponzi language.

Month 3 — Life Simulator: The Goal Planner shows Priya: “If you save ₹3,000/month in an index fund SIP, your daughter's engineering education (₹8 lakh in 12 years) is 92% funded. Here's the comparison with your current FD.”

Month 6 — Behavioral Change: Priya has maintained a 180-day savings streak. The Behavior Engine celebrates: “Priya, you've saved ₹18,000 in 6 months! You're in the top 15% of savers in your category.” She feels confident and in control.

9.2 Rajesh’s Journey

Rajesh’s ArthRakshak Experience

Day 1 — Income Pattern Detection: Rajesh connects his earnings data. The Cashflow Agent identifies his income pattern: high-earning weekends, low-earning weekdays, with monthly variance of 40%. The Twin is built.

Week 2 — Safe-to-Spend Alert: “Rajesh, based on your earnings this week (₹ 4,200) and your rent due on the 5th (₹ 6,000), you’re safe to spend ₹ 1,800 this week. I’ve set aside the rest.”

Month 1 — Debt Trap Warning: The Twin detects a pattern: Rajesh’s spending in the last 5 days exceeds his projected income. Proactive alert: “Your current pace puts you ₹ 3,200 short by month-end. Here are 3 ways to adjust.”

Month 2 — Emergency Fund: The Goal Agent proposes: “Saving ₹ 50/day on your good-earning days builds a ₹ 9,000 emergency fund in 6 months. That’s 15 days of survival buffer.”

Month 4 — Scam Prevention: Rajesh almost downloads a loan app offering “instant ₹ 20,000.” The Scam Shield intercepts: **High Risk** — APR is 84%, not in RBI’s DLA Directory, 47 reported complaints. Rajesh is saved from a debt spiral.

10 Impact Measurement & Sustainability

10.1 Key Performance Indicators

Table 9: Impact Metrics Framework

Metric Category	Specific KPIs	Target (Year 1)
Financial Literacy	Financial literacy score improvement (pre/post)	+30% avg.
Behavioral Change	Users with active savings streak > 90 days	40% of users
Scam Prevention	Scam attempts flagged and blocked	10,000+
Debt Reduction	Users who avoided new high-interest debt	60% of at-risk
Goal Achievement	Users on track for stated financial goals	50%
Inclusion	Users from low-income / rural / disabled segments	45% of base
Trust	Net Promoter Score (NPS)	> 70
Engagement	Monthly Active Users retained at 6 months	55%

10.2 Scalability Path

- Phase 1 (Months 1–6):** Mumbai/Thane pilot with 1,000 users across all four persona types. Validate core agents, tune Digital Twin models, establish NGO partnerships.
- Phase 2 (Months 6–12):** Expand to Maharashtra and Karnataka. Add 5 languages. Integrate WhatsApp Business API for mass reach. Target 50,000 users.
- Phase 3 (Year 2):** National rollout. Partner with Jan Dhan scheme, PMJDY, and self-help groups (SHGs). Target 500,000 users. Add government scheme navigation for 200+ central and state schemes.
- Phase 4 (Year 3+):** Open-source the agent framework for NGOs and educators. Create a “ArthRakshak for Schools” module. Target 5 million users through B2G and B2B2C channels.

10.3 Sustainability Model

Table 10: Revenue Sustainability (Non-Exploitative)

Revenue Stream	Description
B2G Partnerships	Government digital literacy programs, financial inclusion missions
NGO/CSR Funding	CSR mandates from financial institutions for financial literacy
Freemium Model	Free core features; premium Life-Event Simulator and advanced analytics
API Licensing	Scam Shield API licensed to banks, fintech companies, telecom operators
Data Insights	Anonymized, aggregated financial behavior insights (never individual data)

11 Competitive Differentiation

Table 11: ArthRakshak vs. Existing Solutions

Capability	ArthRakshak	Budget Apps	Bank Advisors	AI Chatbots	Fintech Apps
Financial Digital Twin	✓	✗	✗	✗	✗
Irregular Income Support	✓	✗	✗	✗	~
Life-Event Simulation	✓	✗	✗	✗	✗
Scam Shield (Real-time)	✓	✗	✗	✗	~
Behavioral Nudges	✓	~	✗	✗	~
Multilingual (8+ langs)	✓	✗	~	~	✗
Voice-First Interaction	✓	✗	✗	~	✗
Accessibility (Disability)	✓	✗	✗	✗	✗
Human Escalation	✓	✗	✓	✗	✗
Proactive Alerts	✓	~	✗	✗	~
Community Trust Layer	✓	✗	✗	✗	✗

12 Innovation Summary

✓ What Makes ArthRakshak Unique

1. **Financial Digital Twin** — No existing consumer product builds a living, evolving financial model that enables proactive guidance. This concept, borrowed from industrial IoT and adapted for personal finance, is our foundational innovation.

2. **Life-Event Simulator** — Transforms financial literacy from passive information consumption into active experiential learning with the user’s own data. Grounded in behavioral economics research.

3. **Scam Shield as Core Pillar** — Elevated from a feature to a central architecture component with real-time threat detection, risk scoring, and human escalation — directly addressing

₹48,021 crore in annual fraud losses.

4. **Behavioral Finance Integration** — Not just “nudges” — a systematic application of commitment devices, loss framing, social proof, and micro-goals grounded in Thaler’s nudge theory and Kahneman’s prospect theory.
5. **Four-Persona Universal Design** — Every architectural decision is validated against Priya, Rajesh, Kisan, and Divya — ensuring the system works for salaried professionals, gig workers, farmers, and persons with disabilities.
6. **Human-AI-Community Loop** — Acknowledges that financial trust requires human involvement. Integrates NGO verification, community ambassadors, and human escalation as first-class architectural components.
7. **Hierarchical Multi-Agent Architecture** — Goes beyond linear chatbot pipelines to a LangGraph-based stateful orchestration with 8+ specialized agents, persistent memory, and full audit trails.

13 About the Team

Abhisheksingh Dhangar

Full-Stack & AI Engineer

IIIT Pune (Indian Institute of Information Technology, Pune)

Technical Expertise:

- PyTorch, Deep Learning (ViT, YOLO)
- OpenCV, Hugging Face, NLP
- LLMs & GenAI (Gemini API)
- React.js, Redux, Node.js
- FastAPI, Express.js, JWT
- PostgreSQL, MongoDB, SQL
- Docker, Git, CI/CD, CUDA
- C++, Python, JavaScript, Java

Relevant Experience:

- Designed and built Sentinel Core, an AI-powered forensics engine integrating 6 ML pipelines (YOLOv8, ViT, SigLIP)
- Developed full-stack features, optimized SQL schemas and query performance for agri-tech systems at Kisan-365
- Engineered ProofCV AI (resume optimizer) and HireLive (interview platform with video calls) full-stack platforms
- Integrated Gemini LLM APIs for automated ATS matching, document analysis, and dynamic report generation

Motivation:

“Integrating state-of-the-art Deep Learning and GenAI pipelines ensures ArthRakshak acts as a highly intelligent, proactive financial guardian. By building systems that adapt to real-world data and user realities, we can bridge the understanding gap for those who need it most.”

Mohd Bilal Khan

AI Engineer & Backend Developer

IIIT Pune (Indian Institute of Information Technology, Pune)

Technical Expertise:

- Python & FastAPI
- LangChain & LangGraph
- Multi-Agent System Design
- Claude / GPT API Integration
- PostgreSQL & SQLAlchemy
- ChromaDB & RAG Systems
- React & TailwindCSS
- Docker & CI/CD

Relevant Experience:

- Designed and built multi-agent AI systems with LangGraph orchestration
- Backend API development with FastAPI for production-grade applications
- RAG pipeline implementation for domain-specific knowledge retrieval
- Experience with Indian language NLP and speech-to-text systems

Motivation:

“I’ve seen how financial confusion affects families around me — from relatives who bought insurance they didn’t need, to friends trapped in predatory loan cycles. ArthRakshak is built from real frustration with a system that digitizes transactions but not understanding. Every Indian household deserves a financial guardian that speaks their language and understands their life.”

14 References & Data Sources

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ArthRakshak

A Financial Digital Twin for Every Indian Household

Built with care for Priya, Rajesh, Kisan, and Divya — and millions like them.
